

Chlorophyll fluorescence and PAM fluorometry in microalgae

Precaution:

- Stir regularly the sample with an inoculation loop
- Fill three 1ml cuvette with algae. Keep one for tests (sample 0), and two in the dark using aluminium foil (samples 1 and 2). Ensure the sample is free of air bubbles.

Step 1 (~10 minutes): Learn how the machine works

Read the documentation briefly: 1 group could read parts 4-7 and 9 (pages 13-24, 44-45); 1 group could read part 11 (pages 48-60, 63-66).

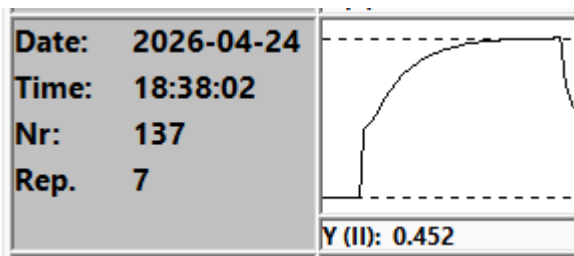
Step 2 (~30 minutes): Test setting-up the PAM controller

Note: Prepare one table sheet to store the data you are going to collect later.

Using the PAM controller, do AUTO-ZERO (no sample) and check if fluo (with sample 0) is in the 200-500 range. If not change PM-GAIN.

Check that ML, and SAT are set to their default values. Change AL and see. If time allows, try to change different SAT-WIDTH and check that time is enough to reach saturation. An example with 1s is given below.

Check how to collect data (MEM).



Step 3 (~45 minutes): Perform an induction curve from dark-adapted cells (sample 1)

Pick one AL of your choice.

Once ready, setup sample 1, making sure to limit its exposure to light. ML and AL should be off at this stage. In case, you could leave the sample 10 minutes in the dark before starting the experiment.

Run an induction curve (during ~20 minutes). Take F' -values on-the-fly. Remember to stir from time to time (not during SAT!).

At the end, collect F_0 , F_M and F'_M stored on the controller. Plot $F'(t)$ and $F'_m(t)$.

How long does it take for the cell to acclimate, meaning for F'_m to converge? The next group will use this as a criterion for the next experiment (step 4).

Estimate and plot the maximal photosynthetic rates Φ_{PSII}^{max} and $\Phi_{PSII}^{max}(I_{AL})$. Comment.

Step 4 (~ 45 minutes): Perform a light curve from dark-adapted cells (sample 2).

Similar procedure than step 3. The goal here is to get $\Phi_{PSII}^{max}(I)$, for different light intensities. It could be done "manually" or using the LIGHT CURVE mode.

Take F' -values on-the-fly. Remember to stir from time to time (not during SAT!).

At the end, collect F_0 , F_M and F'_M stored on the controller. Plot $F'(t)$ and $F'_m(t)$. Estimate and plot the maximal photosynthetic rates $\Phi_{PSII}^{max}(I)$. Comment.